

Research on Document Image Binarization: A Survey

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Abstract—In the digital age, the majority of paper documents are converted into electronic formats for preservation purposes. However, the conversion process can result in significant degradation of the document images, which can subsequently lead to difficulties in reading, text recognition, and other digitisation processes. Document image binarisation is a crucial pre-processing step for automatic document recognition and data analysis. The effectiveness of this binarisation directly impacts the accuracy of the digitisation operation. This paper presents a summary of the research results based on traditional thresholding methods, image feature methods and some learning methods, as well as the evaluation indexes of binarisation performance. It also discusses the characteristics and limitations of these methods. Finally, it presents a summary of relevant methods for document image binarisation and points out several promising research directions.

Index Terms—Degraded document images, Document binarization, Thresholding, Machine learning, Deep learning.

I. INTRODUCTION

THE text is the carrier of human civilisation. Prior to the digital era, text was typically handwritten or printed on paper. With the continuous development and wide application of computer technology, the way people save text has gradually shifted from paper documents to digital documents. The process of storing these documents over a long period of time can cause their document quality to begin to deteriorate. In order to maintain the quality of the original content, it is therefore necessary to convert these documents into digital documents. Binarisation is a key step in this process. Technically, document image binarisation is the process of converting a colour or greyscale document image into a black and white version only. The binarisation result is considered successful when it retains meaningful information and discards noisy information.

The development of computer vision technology has led to advances in the field of document image binarisation. Initially, this technology was developed from threshold-based binarisation methods and image feature-based methods, which were subsequently replaced by automatic machine learning methods. The diversity of the types of degraded documents and the difficulty of defining the diverse features of these documents by threshold-based binarisation methods and image feature-based methods have led to the current mainstream

method of using deep learning to binarise degraded document images.

The paper is divided into four sections. The second section presents the challenges encountered in the task of document binarisation. The third section reviews representative binarisation methods. Finally, in the fourth section, the full paper is summarised with an outlook.

II. CHALLENGES IN DOCUMENT BINARIZATION TASKS

1) *Uneven illumination of the document.*: This is due to the superimposition of a background. This causes artefacts to appear in the process of document image binarisation, which in turn leads to the extraction of erroneous text from the artefacted areas.

2) *Contrast variation.*: Contrast is the difference between objects in an image with high-intensity pixels and low-intensity pixels, which can also be defined as a change in brightness. This can lead to difficulties in the document image binarisation process when applying traditional threshold-based methods to separate the foreground text from the background of the document image.

3) *Paper ink bleed and plaque.*: Due to the disparity in paper quality and ink quality, bleed-through occurs when text is written on both sides of the paper, resulting in the appearance of ink on the reverse side. This hinders the differentiation of the patches from the characters in the binarisation process.

4) *Character fading.*: Character fading is a phenomenon whereby the text image undergoes a gradual loss of colour and clarity over an extended period of time, due to the influence of external factors. This results in the edges of the characters becoming fuzzy and unclear, which in turn affects the accuracy of character recognition.

5) *Paper degradation.*: In the event of paper deterioration, such as yellowing, brittleness, and so forth, the colour, size, shape, and so on, of the characters on the image may undergo change, thereby affecting the outcome of binarisation.

III. DEGRADED DOCUMENT IMAGE BINARIZATION METHOD

Document image binarisation is the process of setting the grey value of a pixel point on a document image to either

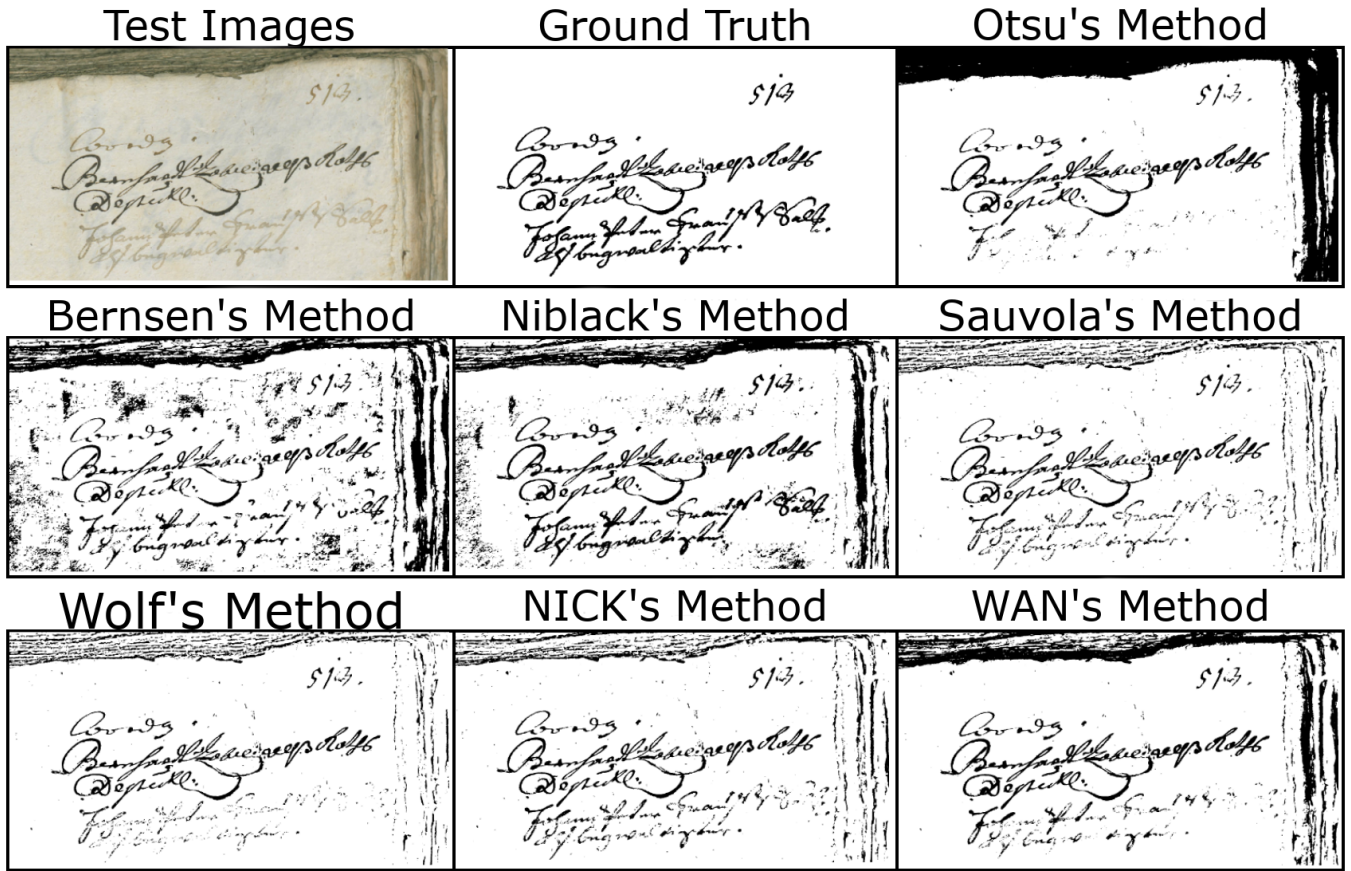


Fig. 1: Threshold-based binarization results for degraded document images.

0 or 255. A grey value of 0 represents the pixel value of the white background, while a grey value of 255 represents the pixel value of the black text in the foreground. This section examines thresholding-based methods, image feature-based methods, traditional machine learning, and deep learning methods. It also summarises their respective features and limitations.

A. Threshold-based binarization method

In the early binarization methods the pixels of the degraded document image are grouped by introducing a threshold value. The threshold value $T(i, j)$ is a two-dimensional function defined on the domain $I(i, j)$ of the original image, which determines the binarization threshold of the corresponding pixel of the original image. Therefore, the pixels of a binarized image are defined by the following equation:

$$B(i, j) = \begin{cases} 0 & \text{if } I(i, j) < T(i, j) \\ 255 & \text{if } I(i, j) \geq T(i, j) \end{cases} \quad (1)$$

Thresholding based binarisation methods can be divided into global thresholding method and local thresholding method, global thresholding method is sufficient to consider a single threshold only for the document image. In global thresholding method thresholds are generally determined by grey scale

histogram analysis and image entropy based analysis. The obvious advantage of the global thresholding method is that it is extremely fast, however, one of the major disadvantages of the global thresholding method is that it suffers a lot when the degradation of the image is only present locally. The local thresholding method considers a different threshold for each pixel by considering the grey values of the neighbouring pixels. The local thresholding method compensates to some extent for the shortcomings of the global thresholding method in dealing with non-uniformly varying documents. However, this type of method selects a different threshold for each pixel by the statistical information of the central pixel's neighbourhood window, and the computational complexity changes with the size of the neighbourhood. And the local thresholding method needs to traverse each pixel, so the computational efficiency of the local thresholding method will be much lower than the global thresholding method [1]. Threshold-based binarization methods are summarized in Table I.

B. Binarization method based on image features

Some methods based on edge detection have also been employed in document image binarisation, where information about stroke and background pixel intensities is estimated as the main driver of active contour propagation (level set function displacement) by using a local linear model. Binarisation

TABLE I: Table of traditional binarization methods

Algorithm classification	Algorithm name	Year of publication	Algorithm Description
Global Thresholds	Otsu [2]	1979	Select the gray value corresponding to the maximum interclass variance in the histogram as the global threshold value
	Kapur [3]	1985	Maximization formula based on image entropy
	Kittler [4]	1986	A Minimum Error Thresholding Method with Improvements to the Bayesian Minimum Error Thresholding Criterion
	Li [5]	1998	Minimum cross-entropy thresholding method based on one-point iterative method
	Mello [6]	2000	Determining global thresholds using Shannon entropy
	Silva [7]	2006	Global thresholding using entropy of image histograms
Local Thresholds	Niblack [8]	1985	Adaptive thresholding method for calculating the mean and variance in the neighborhood window of each pixel
	Bernsen [9]	1986	Calculate the average of the minimum and maximum values of the contrast in the neighboring window pixels to determine the local thresholds
	Sauvola [10]	2000	Improvement of the Niblack method by using the dynamic range of the standard deviation of the image gray values
	Wolf [11]	2002	Global statistical normalization based on the Sauvola method
Edge-Based	Lu [12]	2010	Binarization Method Using Background Surface and Text Stroke Edge Information
	Rivest [13]	2012	Local linear probability model based on embedding in level sets
	Su-Lu [14]	2012	Binarization by combining contrast mapping with Canny edge mapping
	Lelore [15]	2013	Based on a dual-threshold edge detection method
	Hadjadj [16]	2017	Initializing active contour models using contrast edges
Image Processing	Nafchi [17]	2014	Extraction of main features by phase and use of adaptive Gaussian and median filters
	Zemouri [18]	2014	Improvement of image quality by contour wave transform and thus selection of important coefficients
	Michalak [19]	2019	Segment the image into chunks to eliminate non-uniform background and apply Otsu's method for binarization
	Sehad [20]	2019	Extraction of texture information features of an image using Gabor filter
Markov Random Field	Lelore [21]	2009	A Markov Random Field Model for Binarization
	Lettner [22]	2009	The Gibbs distribution is defined in a different form
	Kuk [23]	2009	A variant of the Gibbs model
	Peng [24]	2010	Introducing a new edge potential feature in the Markov random field framework
	Su [25]	2012	Combining Markov Random Fields and Edge Information and Classifying Uncertain Pixels Using Foreground and Background Pixels
	Howe [26]	2013	Global Energy Function Labeled Binarized Pixels Inspired by Markov Random Field Models
Shallow Machine Learning	Hamza [27]	2005	Self-organizing mapping (SOM) extracts its most representative gray level or color
	Rabelo [28]	2011	A local mean-based thresholding method for complex background documents in multilayer perceptron machines is proposed
	Pastor [29]	2013	Using F-Measure as a Loss Function for Backpropagation
	Kefali [30]	2014	Front and rear view separation based on local intensity and global statistical feature MLPs
Supervised Learning	Chou [31]	2010	Region binarization using support vector machines to construct decision functions from information provided by training samples
	Chattopadhyay [32]	2013	Selecting the Best Binarization Algorithm Using Support Vector Machines
	Wu [33]	2016	Learning more complex decision functions based on multiple feature sets
	Westphal [34]	2018	Howe's approach based on heterogeneous computing platforms
	Xiong [35]	2018	Select an optimal global threshold for each image block using a support vector machine
Clustering	GiB [36]	2018	Designed at pixel level to extract clustering features based on game theory
	Gattal [37]	2018	A method for clustering K-means based on automatic tuning of parameters is proposed
	Krantz [38]	2019	A clustering-based sample selection method is proposed
	Bera [39]	2021	Based on the integration of three classical clustering methods (fuzzy C-means, K-medoids and K-means++)
RNN	Afzal [40]	2015	Proposing the use of Long Short-Term Memory (LSTM) specialized in ultra-long sequences to solve document image binarization
	Westphal [41]	2018	Proposing an RNN-based binarization method and processing multidimensional inputs using Grid LSTM units
CNN	Pastor [42]	2015	First to propose binarization with CNNs
	Tensmeyer [43]	2017	Proposing a FCN combining P-FM and FM loss training
	DeepOtsu [44]	2019	Used U-Net to learn degraded document images after binarization using Otsu thresholding
	Huang [45]	2020	Combined local and global U-Net approach
	DP-LinkNet [46]	2021	D-LinkNet with Hybrid Null Convolution and Spatial Pyramid Pool Modules
GAN	Bhunja [47]	2019	D-LinkNet with Hybrid Null Convolution and Spatial Pyramid Pool Modules
	Tensmeyer [48]	2019	Proposing a method to generate realistic synthetic data using CycleGAN
	Dang [49]	2021	GAN-based stroke preservation method
	Kumar [50]	2021	A joint discriminator is proposed
	Tamrin [51]	2021	A two-stage unsupervised model based on GAN and U-Net
	Jemni [52]	2022	A GAN method with integrated handwritten text recognizer
Transformers	DocEnTr [53]	2022	First to use Vision Transformers for document image binarization

using edge detection suffers from the problem that the detected edges do not completely enclose the foreground region. Some scholars have also used image processing methods in the task of document image binarisation, and since some of the image processing based methods are combined with traditional thresholding methods, these methods are able to ameliorate to some extent the problem of high sensitivity of traditional thresholding based methods to the document conditions. The following summarizes the binarization methods based on image features are grouped in Table I.

C. Machine learning based binarization methods

In general, adaptive thresholding based binarisation cannot avoid noise in the background as well as voids in the text characters, while Markov Random Field (MRF) [54] based image restoration methods have attracted the interest of researchers in the field of document processing. Multilayer Perceptron (MLP) is a feed-forward structure. It has a strong ability to learn complex data, but the model is only able to learn very simple features. Support Vector Machine (SVM) is a supervised machine learning algorithm. It can be used to classify blocks of images. And select the best threshold for binarisation. The background and foreground pixels are roughly estimated using clustering method. These estimated pixels are used as a priori values for implementing GraphCut source and sink. The summary of machine learning based binarization methods are summarized in Table I.

D. Deep learning based binarization method

In recent years, deep learning has also gradually become a mainstream method for document image binarisation processing. In deep learning frameworks recurrent neural network (RNN) uses recursive structure to process sequence data [55], [56]; fully connected network (FCN) is a neural network consisting of multiple linear layers and nonlinear activation function superposition [57], [58]; convolutional neural network (CNN) introduces convolutional and pooling layers to deal with displacement invariant data [59], [60]; generative adversarial network (GAN) [61] provides a method to learn deep representations without the need for large amounts of labelled training data; and Transformer is a new type of neural network, which mainly exploits the self-attention mechanism [62], [63] to extract image intrinsic features [64]. And it shows excellent application potential in document image binarisation tasks.

IV. SUMMARY

A. Document image binarization algorithm summary

Document image binarization has evolved from traditional threshold-based methods to today's traditional machine learning methods and deep learning methods for more than forty years, and some important results have been achieved.

Therefore, this paper introduces some highly relevant techniques for binarisation of degraded document images and provides a systematic classification as well as a comparative analysis of these binarisation methods. Due to the diversity

of degraded document types, traditional binarization methods based on threshold have poor generalization performance and high sensitivity to document conditions. They usually cannot recover highly degraded images, it is difficult to define the diversity of their features, and they fail to achieve the expected binarization results for different types of degraded documents. Only satisfactory binarization results can be obtained on some specific types of degenerate documents. Subsequently, some scholars have reduced the problem of high sensitivity of traditional thresholding methods by using some features of document images combined with thresholding-based methods. Machine learning-based binarisation methods are more robust than traditional threshold-based and image feature-based methods. It is usually possible to learn more flexible features or decision functions. These features or decision functions are more suitable for image binarisation tasks. However the large labeled dataset required for training. For how to select the appropriate binarisation technique to annotate document images is also one of the challenges.

With the rapid development of computer hardware, the task of using deep learning methods to deal with document binarization has dominated the subsequent timeline, and despite many breakthroughs in recent years, deep learning is often considered a convoluted black-box learning procedure. The approach differs from traditional methods with clear physical and mathematical meaning in that deep learning-based methods extract high-level features that are difficult to interpret. As described in Section 3, many deep learning-based methods have been successively proposed as a way to design an interpretable network where each module gives a specific operation. The combination of model-driven and data-driven approaches provides a new perspective for understanding the workflow in deep learning black-box networks.

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